

Bookish - Full Stack Web Development Project

Group ID: 53

Course: Framework Driven Full Stack Development

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Project Overview

Bookish is a unified web platform designed for book buyers, sellers, and administrators. It merges the core features of popular online bookstores and e-reading services into a single integrated system. The platform provides an engaging and efficient user experience by offering role-based access and functionality across all user types. Buyers can browse, subscribe, and maintain their personal libraries; sellers can manage book uploads, inventory, and orders; and administrators oversee user management, content moderation, and reporting. By bridging the gap between book e-commerce and digital reading, Bookish aims to deliver a comprehensive solution for book enthusiasts and vendors alike.

Business Model

Bookish follows a typical e-commerce revenue flow similar to major online marketplaces like Amazon or Flipkart. When a buyer purchases a book, the payment is initially processed through a secure payment gateway and held in the admin account. Once the order is delivered and confirmed by the buyer, the admin deducts a small commission as a service charge and transfers the remaining amount to the seller's account. This ensures transaction security, transparency, and proper revenue distribution between the platform and the sellers. The same flow applies for subscription plans and paperback orders, ensuring reliability across all types of transactions.

Objectives and Problem Statement

The primary objective of Bookish is to create a full-stack web solution that seamlessly integrates the processes of buying, selling, and reading books within one application. The project addresses the fragmentation present in existing platforms by offering unified access for all stakeholders—buyers, sellers, and administrators. The problem lies in managing multiple systems for commerce, user interaction, and reading experiences. Bookish resolves this by consolidating these modules under a single interface with dynamic, asynchronous operations.

System Roles and Functionalities

Buyer:

- Browse books and add them to cart or personal library.
- Manage subscriptions with three available plans unlocking full access.
- Track reading progress and maintain daily streaks.
- Upload and view community video reviews.
- Submit and track complaints via a dedicated dashboard.
- Manage personal profile, orders, and addresses.

Seller:

- Upload books with metadata, manage inventory, and track orders.
- Access a dedicated dashboard showing sales data and top-selling books.
- Submit complaints or queries to administrators.

Administrator:

- Oversee all user accounts, roles, and verifications.
- Moderate uploaded book content (approve/reject functionality).
- Manage and monitor orders with analytics and reporting dashboards.
- Handle complaint resolution efficiently through an admin panel.

Individual Contributions

Ayush Pratap Singh: Implemented payment gateway integration (subscriptions & paperback purchases), video review feed, buyer's library, admin dashboard, and system reports.

Piyush Kumar: Developed buyer's profile page with edit functionality, homepage with search and filter features, seller upload book interface, and admin user management.

Ujjwal Singh: Order tracking system for all roles, seller dashboard with analytics, and admin content moderation.

Daivik Wadhwani: Complaint registration and resolution for all roles, cart functionality for buyers, and seller inventory page displaying book approval status.

Gugulothu Nithin: Implemented login and register pages, logout functionality, contact and about pages, and designed responsive headers and footers.

Technology Stack Frontend

Frontend: HTML/EJS, CSS, Tailwind, JavaScript

Backend: Node.js with Express.js

Database: MongoDB (using Mongoose ODM)

Other Tools: Sessions for authentication, Stripe for payment integration.

Architecture Overview

The Bookish application follows a Model–View–Controller (MVC) architecture. The backend, powered by Node.js and Express, handles routing, authentication, and database interactions through Mongoose models. The frontend uses EJS templating along with Tailwind CSS and JavaScript to provide dynamic, responsive, and asynchronous user experiences. Fetch APIs are used to dynamically update content such as book lists, carts, and dashboards without reloading the page. MongoDB serves as the primary database, storing structured collections for users, books, orders, and complaints. This design ensures scalability, modularity, and responsiveness across user roles.

Demonstration and Evidence

The project demonstration video showcases the major implemented features, including validation, dynamic content rendering, and asynchronous API calls across all user roles. The video provides a live walkthrough of the platform's UI and backend communication flow. Demo Video Link:

<https://www.youtube.com/watch?v=0gduV02pDyA>

Additionally, supplementary evidence such as network request screenshots, Git logs, and presentation slides are maintained in the project repository. These artifacts demonstrate successful implementation, testing, and version control practices throughout development.

Future Scope

Future enhancements could include mobile optimization, AI-driven book recommendations, and a user-rating system. With these expansions, Bookish can evolve into a complete ecosystem for digital reading and publishing.

Conclusion

Bookish successfully integrates the diverse functionalities of online book commerce and digital reading platforms. The system demonstrates a well-structured full-stack implementation with robust backend services, interactive frontend components, and role-based features.